

Idempotency: Make retries safe.

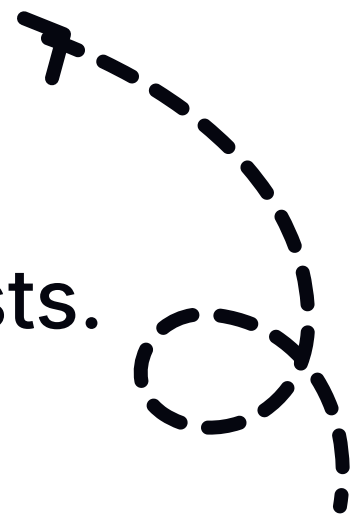
Networks fail. Clients retry.
Your system should be ready
for **both.**



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One customer. **Three** charges.

The client never sent three requests.
It sent one — and retried twice.



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03. WHAT ACTUALLY HAPPENED 3/9

The response died, not the write

`POST /payments → 201 • response lost`

Your server committed the charge. The connection dropped before the response came back. The client saw a timeout — and retried.

01 Attempt 1 — charge committed, response lost

02 Attempt 2 — retry, charged again

03 Attempt 3 — retry, charged again



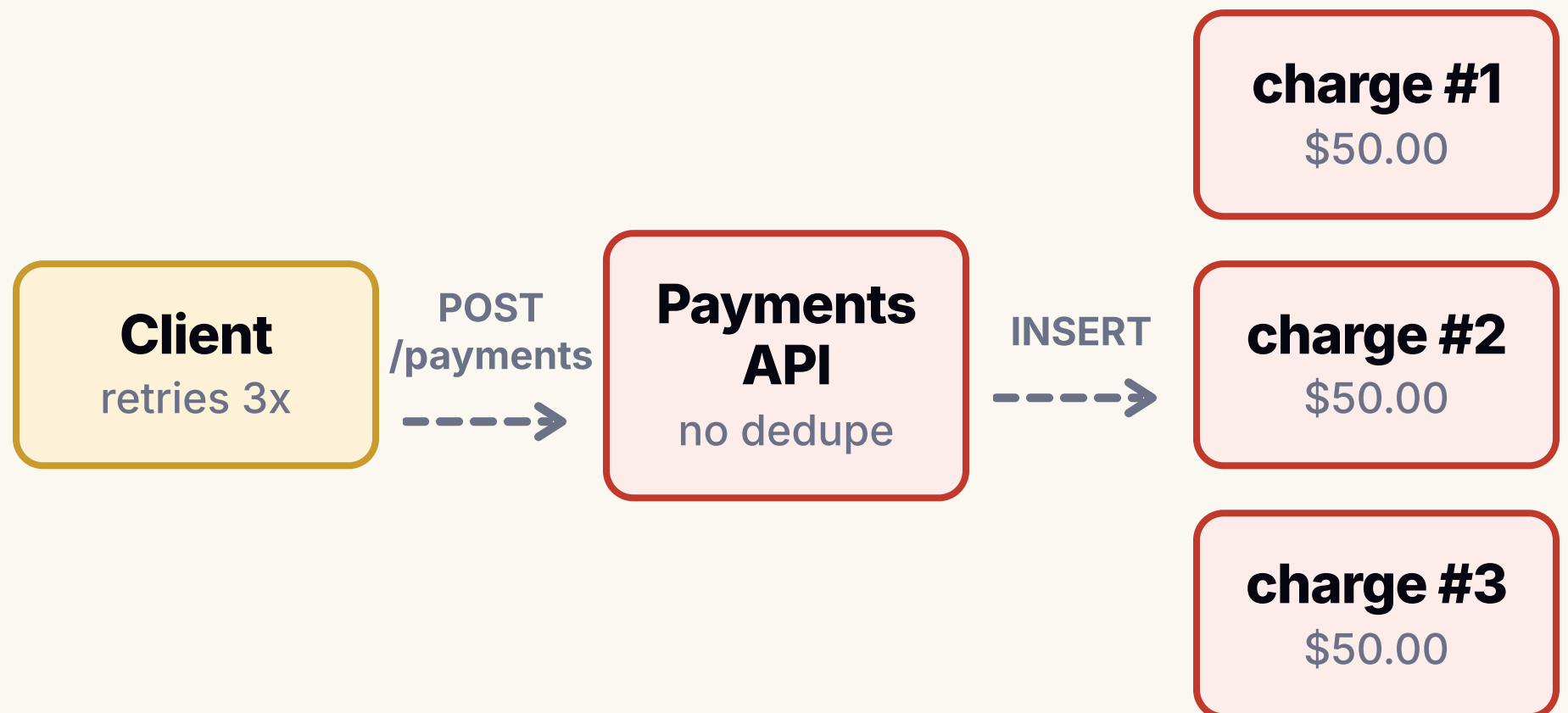
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The retry storm

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A timeout is not a failure.



A timeout tells you the response is missing. It never tells you the write didn't happen.



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05. THE FIX

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Give every attempt a name

One key per **intent** — not per attempt. The same key rides every retry.

WITHOUT A KEY

POST /payments



Every retry looks like a brand-new order.

WITH A KEY

Idempotency-Key: 7c9e-4f21



Every retry resolves to the same order.



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06. THE BUG IN EVERY TUTORIAL

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The loser has nothing to replay

20 concurrent retries → 19 saw 'in_progress'

The winner hasn't finished yet, so no response is stored. Returning **200** here tells the client the payment succeeded — before it has.

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- 01 Replay **only** when the stored status is `done`
 - 02 **409 Conflict** while the winner is still in flight
 - 03 This is what **Stripe** returns in the same case



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Claim the key first

```
payments.controller.js JS

const claim = await db.query(
  `INSERT INTO idempotency_keys (key, status)
  VALUES ($1, 'in_progress')
  ON CONFLICT (key) DO NOTHING
  RETURNING key`, [key]);

if (!claim.rowCount) {
  const prior = await load(key);
  // Winner still running – nothing to replay yet.
  if (prior.status === 'in_progress')
    return res.status(409);
  return res.status(prior.code).json(prior.body);
}
```

→ The unique index arbitrates, not an if-statement. Claim it in its own transaction — never the one doing the charge.



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Two retries, one row

schema.sql

SQL

```
CREATE TABLE idempotency_keys (  
  key          TEXT PRIMARY KEY,  
  status       TEXT NOT NULL,  
  response     JSONB,  
  created_at   TIMESTAMPTZ DEFAULT now()  
);
```

```
-- Two retries in the same millisecond both hit  
-- this index. Exactly one INSERT survives.
```

→ No lock. No queue. No distributed coordinator. One **PRIMARY KEY** does the arbitration for you.



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Takeaway

**A retry is not
a new request.
It is the **same intent**,
asking again.**

Design every write endpoint so the second attempt is boring.

Backend Engineering & System Architecture — daily.
Follow @peter-kekpe for the next one.



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